

```
# ===== # TUGAS 01
EKSTRAKSI INFORMASI - SCRAPING DIREKTORI PUTUSAN MA # Tahap 1: Pengambilan Daftar URL Putusan
(Pengadilan Negeri Blitar) ## Deskripsi: # Script ini mengambil kumpulan link putusan pidana umum dari PN Blitar # secara
bertahap dari halaman 1 sampai halaman 10. # Output berupa file teks 'hasilListURLPage.txt' yang berisi seluruh URL putusan. #
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```
In [21]: # Import library yang dibutuhkan
import undetected_chromedriver as uc # Pengendali browser Chrome & bypass proteksi
from bs4 import BeautifulSoup      # Parsing dan pencarian elemen dalam strukt
import time                        # Jeda waktu (delay) saat memuat halaman we
import os                          # Operasi file dan path sistem
import warnings                    # Menyaring peringatan yang tidak krusial
```

```
In [23]: # Inisialisasi Chrome driver
# Browser dibuka satu kali di awal agar hemat memori
driver = uc.Chrome(version_main=153)
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```
In [24]: def getURLfromWeb(url):
    print(f"Mengakses: {url}")
    driver.get(url)

    # Jeda 5 detik agar Cloudflare challenge selesai diverifikasi dan halaman termu
    time.sleep(5)

    # Ambil source HTML halaman yang sudah selesai dirender oleh browser
    html = driver.page_source
    soup = BeautifulSoup(html, 'html.parser')

    # Cari semua tag anchor <a> yang memiliki tautan href
    result1 = soup.find_all('a', href=True)

    urlHasil = []
    for eachResult1 in result1:
        href = eachResult1['href']
        # Filter: Hanya ambil link yang menuju ke halaman detail putusan Mahkamah A
        if 'https://putusan3.mahkamahagung.go.id/direktori/putusan/' in href:
            if href not in urlHasil:
                urlHasil.append(href)

    print(f"Ditemukan {len(urlHasil)} link putusan\n")
    return urlHasil
```

```
In [25]: def main():
    warnings.filterwarnings('ignore')

    # URL awal (halaman 1) dan pola URL untuk halaman selanjutnya (page/2, page/3,
    url1 = 'https://putusan3.mahkamahagung.go.id/direktori/index/pengadilan/pn-blit
    url2 = 'https://putusan3.mahkamahagung.go.id/direktori/index/pengadilan/pn-blit

    listHasil = []
    ulang = 10 # Mengambil 10 halaman daftar putusan

    startTime = time.time()

    # Buka file hasilListURLPage.txt untuk menyimpan kumpulan link putusan
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file_hasil = open("hasilListURLPage.txt", "w", encoding='UTF8')

for i in range(ulang):
    if i == 0:
        url = url1
        print("=== Halaman 1 ===")
    else:
        n = i + 1
        url = url2 + str(n) + '.html'
        print(f"=== Halaman {n} ===")

    # Ambil link putusan dari halaman aktif
    listHasil = getURLfromWeb(url)

    # Tulis link ke file teks
    for listURL in listHasil:
        file_hasil.write(listURL + "\n")

file_hasil.close()
endTime = time.time()

print('Semua halaman berhasil diambil!')
print('Total waktu proses : ', round(endTime - startTime, 2), ' Detik')

main()

```

=== Halaman 1 ===

Mengakses: <https://putusan3.mahkamahagung.go.id/direktori/index/pengadilan/pn-blitar/kategori/pidana-umum-1.html>

Ditemukan 43 link putusan

=== Halaman 2 ===

Mengakses: <https://putusan3.mahkamahagung.go.id/direktori/index/pengadilan/pn-blitar/kategori/pidana-umum-1/page/2.html>

Ditemukan 43 link putusan

=== Halaman 3 ===

Mengakses: <https://putusan3.mahkamahagung.go.id/direktori/index/pengadilan/pn-blitar/kategori/pidana-umum-1/page/3.html>

Ditemukan 43 link putusan

=== Halaman 4 ===

Mengakses: <https://putusan3.mahkamahagung.go.id/direktori/index/pengadilan/pn-blitar/kategori/pidana-umum-1/page/4.html>

Ditemukan 43 link putusan

=== Halaman 5 ===

Mengakses: <https://putusan3.mahkamahagung.go.id/direktori/index/pengadilan/pn-blitar/kategori/pidana-umum-1/page/5.html>

Ditemukan 43 link putusan

=== Halaman 6 ===

Mengakses: <https://putusan3.mahkamahagung.go.id/direktori/index/pengadilan/pn-blitar/kategori/pidana-umum-1/page/6.html>

Ditemukan 43 link putusan

=== Halaman 7 ===

Mengakses: <https://putusan3.mahkamahagung.go.id/direktori/index/pengadilan/pn-blitar/kategori/pidana-umum-1/page/7.html>

Ditemukan 43 link putusan

=== Halaman 8 ===

Mengakses: <https://putusan3.mahkamahagung.go.id/direktori/index/pengadilan/pn-blitar/kategori/pidana-umum-1/page/8.html>

Ditemukan 43 link putusan

=== Halaman 9 ===

Mengakses: <https://putusan3.mahkamahagung.go.id/direktori/index/pengadilan/pn-blitar/kategori/pidana-umum-1/page/9.html>

Ditemukan 43 link putusan

=== Halaman 10 ===

Mengakses: <https://putusan3.mahkamahagung.go.id/direktori/index/pengadilan/pn-blitar/kategori/pidana-umum-1/page/10.html>

Ditemukan 43 link putusan

Semua halaman berhasil diambil!

Total waktu proses : 62.05 Detik

```
In [26]: # Tutup sesi browser setelah proses pengambilan URL selesai  
driver.quit()
```

```
# ===== # TUGAS 01
EKSTRAKSI INFORMASI - SCRAPING DIREKTORI PUTUSAN MA # Tahap 2: Ekstraksi Meta Informasi Putusan & Ekspor
ke File CSV ## Deskripsi: # Script ini membaca daftar URL dari 'hasilListURLPage.txt', kemudian mengakses # halaman detail
setiap putusan untuk mengekstrak informasi perkara: # (Terdakwa, Penuntut Umum, Nomor Putusan, Hakim, Panitera, Amar, dsb).
# Seluruh hasil ekstraksi disimpan ke dalam file CSV 'IE-PN_Blitar-240411100135.csv'. #
=====
```

```
In [19]: # Import library yang dibutuhkan
import undetected_chromedriver as uc # Otomasi browser & bypass proteksi Cloudflar
from bs4 import BeautifulSoup        # Parsing dan ekstraksi struktur DOM HTML
import time                          # Mengatur jeda (delay) render halaman
import os                            # Mengecek keberadaan file di sistem lokal
import warnings                      # Menyaring peringatan yang tidak relevan
import re                            # Regular expression untuk pembersihan tag
import csv                           # Menulis data ke file CSV berformat tabel
```

```
In [20]: # Inisialisasi Chrome driver (versi 153 sesuai Google Chrome di laptop)
# Driver dibuka satu kali di awal agar proses scraping berjalan efisien
driver = uc.Chrome(version_main=153)
```

```
In [21]: def generateFileCSV(listHasil, csvName1):
    csvFolder = "./"
    csvName = csvName1

    # Header kolom sesuai format informasi meta perkara putusan
    header_kolom = (
        "terdakwa", "penuntut_umum", "nomor", "tingkat_proses", "klasifikasi",
        "kata_kunci", "tahun", "tanggal_register", "lembaga_peradilan",
        "jenis_lembaga_peradilan", "hakim_ketua", "hakim_anggota", "panitera",
        "amar", "amar_lainnya", "catatan_amar", "tanggal_musyawarah",
        "tanggal_dibacakan", "kaidah", "abstrak", "url"
    )

    # Gunakan encoding UTF-8 agar karakter khusus (tanda baca, simbol hukum) tidak
    if os.path.exists(csvName):
        f = open(csvName, 'a', newline='', encoding='utf-8')
        print("File CSV sudah ada, menambahkan data...")
        w = csv.writer(f)
    else:
        f = open(csvName, 'w', newline='', encoding='utf-8')
        print("File CSV belum ada, membuat file baru...")
        w = csv.writer(f)
        w.writerow(header_kolom)

    # Tulis setiap baris data putusan ke file CSV
    for s in listHasil:
        w.writerow(s)

    f.close()
    return "\nCreate Csv file Berhasil\n"
```

```
In [22]: def generateMeta(urlMeta):
    url = str(urlMeta).strip()

    # Akses halaman detail putusan dengan browser Chrome
```

```

driver.get(url)
time.sleep(3) # Jeda 3 detik agar Cloudflare memvalidasi dan halaman selesai d

soup = BeautifulSoup(driver.page_source, 'html.parser')
cleanTags = re.compile('<.*?>') # Pola regex untuk membersihkan tag HTML (<...

listMeta = []

# Cari tabel meta info perkara pada tag ul class 'portfolio-meta nobottommargin'
portfolio_meta = soup.find("ul", {"class": "portfolio-meta nobottommargin"})
if not portfolio_meta:
    return []

table = portfolio_meta.find("table")
rowsMETA1 = table.findAll("tr") if table else []
rowsMETA2 = soup.findAll("ul", {"class": "portfolio-meta nobottommargin"})

# Ekstraksi setiap baris pada tabel metadata
for row in rowsMETA1:
    coll = row.findAll("td")

    # Jika ada 2 kolom (Label & Nilai), ambil nilai di kolom kedua
    if len(coll) > 1:
        cleantext2 = (re.sub(cleanTags, '', str(coll[1]))).strip()
        listMeta.append(cleantext2.replace('\n', ' '))
    else:
        # Kolom tunggal biasanya berisi identitas pihak (Terdakwa / Penuntut Um
        cleantext1 = (re.sub(cleanTags, ' ', str(coll[0]))).strip()
        pidorpdtdt = re.search(r'(.*)/Pdt.(*)', str(cleantext1), re.M | re.I)

        # Identifikasi perkara pidana atau perdata
        if pidorpdtdt == None:
            entTerdakwah = re.search(r'(.*)Terdakwa:(.*)', str(cleantext1), re.
            entPenuntut = re.search(r'Penuntut Umum:(.*)Terdakwa:', str(cleante
        else:
            entTerdakwah = re.search(r'(.*)Tergugat:(.*)', str(cleantext1), re.
            entPenuntut = re.search(r'Penggugat:(.*)Tergugat:', str(cleantext1)

        # Ambil nama terdakwa jika pola cocok
        if entTerdakwah == None:
            listMeta.append("")
        else:
            listMeta.append(entTerdakwah.group(2).strip())

        # Ambil nama penuntut umum jika pola cocok
        if entPenuntut == None:
            listMeta.append("")
        else:
            listMeta.append(entPenuntut.group(1).strip())

# Ekstraksi tautan unduh dokumen putusan jika ada
if len(rowsMETA2) > 1:
    urlDL = rowsMETA2[1].findAll("li")
    if len(urlDL) > 4:
        urlDLStr = str(urlDL[4])
        listMeta.append(urlDLStr[urlDLStr.find("https"):urlDLStr.find('>')])

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        else:
            listMeta.append("")
    else:
        listMeta.append("")

    return listMeta

```

```

In [23]: def main():
    warnings.filterwarnings('ignore')

    # Konfigurasi file input dan file output CSV sesuai penamaan tugas (IE-NamaPN-n
    fileListURL = "hasilListURLPage.txt"
    fileMetaCSV = "IE-PN_Blitar-240411100135.csv"
    listHasil = []

    startTime = time.time()

    # Membaca daftar URL putusan yang sudah diambil pada tahap 1
    openfileListURL = open(fileListURL, "r", encoding='UTF8')
    rawListURL = openfileListURL.readlines()
    openfileListURL.close()

    # Filter: Hapus link duplikat dari sidebar agar setiap putusan unik dan hemat w
    bacaListURL = list(dict.fromkeys([u.strip() for u in rawListURL if u.strip()]))
    print(f"Total URL putusan unik yang akan diproses: {len(bacaListURL)}")

    # Iterasi pengambilan metadata dari setiap URL putusan
    i = 1
    for barisURL in bacaListURL:
        try:
            hasil = generateMeta(str(barisURL))
            if hasil:
                listHasil.append(hasil)
                print(f"===== ROW HASIL ===== {i} / {len(bacaListURL)}")
            else:
                print(f"Row {i}: Format tidak sesuai / kosong")
        except Exception as e:
            print(f"Error Get Meta Inf row {i}, {e}")
        i = i + 1

    # Simpan kumpulan metadata yang berhasil diambil ke dalam file CSV
    createFile = generateFileCSV(listHasil, fileMetaCSV)
    print(createFile)

    endTime = time.time()
    print('Semua proses scraping metadata selesai!')
    print('Total waktu pemrosesan : ', round(endTime - startTime, 2), ' Detik')

main()

```

Total URL putusan unik yang akan diproses: 237

===== ROW HASIL ===== 1 / 237

Row 2: Format tidak sesuai / kosong

===== ROW HASIL ===== 3 / 237

===== ROW HASIL ===== 4 / 237

===== ROW HASIL ===== 5 / 237

Row 6: Format tidak sesuai / kosong

Row 7: Format tidak sesuai / kosong

===== ROW HASIL ===== 8 / 237

Row 9: Format tidak sesuai / kosong

===== ROW HASIL ===== 10 / 237

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Row 41: Format tidak sesuai / kosong

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Row 188: Format tidak sesuai / kosong

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File CSV sudah ada, menambahkan data...
```

Create Csv file Berhasil

Semua proses scraping metadata selesai!
Total waktu pemrosesan : 1336.1 Detik

```
In [6]: # Tutup sesi browser setelah proses scraping selesai
        driver.quit()
```